THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 9 — MATHS FULL SYLLABUS PAPER SET 02

Full Syllabus (Chapters 1–14) | RBSE Pattern | Session 2026–27

Time: 3 Hours 15 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 80**

General Instructions:

1. All questions are compulsory.
2. Q. 1–20 are Objective Type (1 mark each).
3. Q. 21–28 are Very Short Answer Type (2 marks each).
4. Q. 29–36 are Short Answer Type (3 marks each).
5. Q. 37–40 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 35 to 40.
7. Write the question number before each answer.
8. Draw neat figures wherever required.

SECTION — A (Objective Type Questions) 20 × 1 = 20

Choose the correct option and write it in your answer book.

1. The three points $O(0,0)$, $A(3,4)$ and $B(-3,-4)$: 1

(a) form a right triangle **(b)** form an equilateral triangle
(c) form an isosceles triangle **(d)** are collinear

2. The value of the linear polynomial $-2x-1$ at $x=5$ is: 1

(a) -11 **(b)** -13 **(c)** -9 **(d)** 11

3. Rational numbers are NOT closed under: 1

(a) division by zero (undefined) **(b)** multiplication
(c) subtraction **(d)** addition

4. A student writes $(a+b)^2 = a^2+b^2$. The correct response is: 1

(a) Correct only for $a>b$ **(b)** Correct for all a, b
(c) Correct for all positive a, b **(d)** Incorrect; the term $2ab$ is missing

5. A chord of length 16 cm is drawn in a circle of radius 17 cm. The distance of the chord from the centre is: 1

(a) 17 cm **(b)** 15 cm **(c)** 25 cm **(d)** 9 cm

6. Taking $\pi = \frac{22}{7}$, the circumference of a circle of radius 21 cm is: 1

(a) 66 cm **(b)** 264 cm **(c)** 132 cm **(d)** 1386 cm

7. As the number of trials increases, the experimental probability tends to get closer to the: 1

(a) theoretical probability **(b)** number of trials
(c) sample space size **(d)** value 1

8. A letter is picked at random from the word ‘MATHEMATICS’. The probability of picking the letter M is: 1

(a) $\frac{3}{11}$ **(b)** $\frac{2}{11}$ **(c)** $\frac{9}{11}$ **(d)** $\frac{1}{6}$

9. The Virahānka–Fibonacci sequence was first written down in connection with: 1

(a) the area of circles **(b)** planetary motion
(c) Prakrit metre and poetry **(d)** calendar design

10. ‘n is a perfect square’ and ‘n has an odd number of factors’ are related as: 1

(a) each implies the other **(b)** only the second implies the first
(c) only the first implies the second **(d)** neither implies the other

11. The converse of the proposition "If it is Sunday, then the school is closed." is: 1

(a) If the school is not closed, then it is not Sunday.
(b) If the school is closed, then it is Sunday.
(c) The school is closed on Sunday.
(d) If it is not Sunday, then the school is not closed.

12. Marks of 70, 80, 60 are combined with weights 2 : 3 : 5. The weighted average is: 1

(a) 69 **(b)** 70 **(c)** 67 **(d)** 68

13. **Assertion (A):** In the addition algorithm, the carry is never more than 1. 1
Reason (R): The sum of two digits and a carry of at most 1 is at most 19.

(a) Both A and R are true and R is the correct explanation of A
(b) Both A and R are true but R is not the correct explanation of A
(c) A is true but R is false
(d) A is false but R is true

14. The diagonals of a quadrilateral ABCD are AC = 16 cm and BD = 12 cm. The perimeter of the parallelogram formed by joining the midpoints of its sides is: 1

(a) 28 cm **(b)** 56 cm **(c)** 14 cm **(d)** 96 cm

15. In a parallelogram ABCD, $\angle A = 110°$. Then $\angle B$ is: 1

(a) 70° **(b)** 35° **(c)** 250° **(d)** 110°

16. For which value of k does the pair kx + 2y = 4 and 3x + 6y = 12 have infinitely many solutions? 1

(a) 6 **(b)** 1 **(c)** 2 **(d)** 3

17. If x and y satisfy 3x + 2y = 14 and x − y = -2, then the value of 2x + y is: 1

(a) 8 **(b)** 11 **(c)** 6 **(d)** 9

18. **Assertion (A):** The volume of a pyramid with base area B and height h is $\frac{1}{3}Bh$. 1
Reason (R): The volume of a cuboid is lwh.

(a) Both A and R are true and R is the correct explanation of A
(b) Both A and R are true but R is not the correct explanation of A
(c) A is true but R is false
(d) A is false but R is true

19. A pyramid has a base of area 48 cm^2 and a height of 9 cm. Its volume is: 1

(a) 216 cm^3 **(b)** 72 cm^3 **(c)** 144 cm^3 **(d)** 432 cm^3

20. The total surface area of a cuboid of length 12 cm, width 5 cm and height 4 cm is: 1

(a) 138 cm^2 **(b)** 128 cm^2 **(c)** 256 cm^2 **(d)** 240 cm^2

SECTION — B (Very Short Answer Type) 8 × 2 = 16

Answer in about 20–30 words each.

21. Is the point $(-5, -12)$ at the same distance from the origin as $(13, 0)$? Justify. 2

22. A chess club charges ₹200 joining fee plus ₹50 per match. Write the expression for the amount paid for m matches and find m if a player paid ₹750. 2

23. A spice trader has a loan of ₹850, makes a profit of ₹1,200 and then a loss of ₹450. Write the sum using integers and find his final standing. 2

24. The area of a parallelogram is 96 cm^2 and its base is 12 cm. Find its height. 2

25. Two equal glasses have 10% and 20% jaggery lemonade. Without calculation, what is the jaggery percentage after mixing? Why? 2

26. Write a short note on the origin of the word ‘algorithm’. 2

27. True or false, with a reason: a rhombus whose diagonals are perpendicular is a square. 2

28. Write the total surface area of a cylinder closed at both ends, and explain each part. 2

SECTION — C (Short Answer Type) — 8 × 3 = 24

Answer in about 40–60 words each.

29. Write the converse of “If $x = y$, then $a + x = a + y$”. Show that both are true and say where this is used. 3

30. Three vertices of a rectangle are $(2,\ 1)$, $(8,\ 1)$ and $(8,\ 5)$. Find the fourth vertex and the length of the diagonal. 3

31. Draw (describe the points for) the graphs of $y = \frac{1}{2}x$, $y = x$ and $y = 2x$ using suitable points. What does the value of a in $y = ax$ tell about the line? 3

32. Show that $1, 3, 6, 10, \ldots$ is neither an AP nor a GP. 3

33. Solve $11x + 9y = 73$ and $9x + 11y = 67$ without using the substitution method directly, by first adding and subtracting the equations. 3

34. ABCD is a parallelogram of area 84 cm^2. Find the area of $\triangle ABD$ and justify. 3

35. A chord of a circle of radius r is at distance d from the centre. Derive the chord length. Hence find it for $r = 10$ and $d = 6$. 3

OR

Find the centre and radius of the circle through $(0, 0)$, $(6, 0)$ and $(0, 8)$. 3

36. **Case study:** Rohan lists divisors(50) = [1, 2, 5, 10, 25, 50] and divisors(75) = [1, 3, 5, 15, 25, 75]. 3

(i) Write the list of common divisors.

(ii) Find gcd(50, 75).

(iii) Why is the gcd the rightmost element?

OR

Verify, for m = 33 and n = 7, that the common divisors of m and n are the same as the common divisors of n and m mod n. 3

SECTION — D (Long Answer Type) — 4 × 5 = 20

Answer in about 100–150 words each.

37. ABCD is a parallelogram with $AB \neq BC$. Prove that the four angle bisectors meet pairwise at the vertices of a rectangle. Why is $AB \neq BC$ assumed? 5

OR

Case study: A tiler wants to tile a floor with identical copies of a non-convex quadrilateral DART with angles $40^\circ,\ 50^\circ,\ 30^\circ$ and 240°. 5

(i) Check the angle sum.

(ii) Four copies are placed around a point using each angle once; what is the angle total?

(iii) Describe Method 1 for adding more tiles.

(iv) Why can the plane not be tiled with regular pentagons?

38. Show that the following propositions are false by giving a counterexample: **5**

(i) If n is an odd number, then $n^2 + 2$ is prime.

(ii) If n is a prime, then $n^2 - 1$ is a multiple of 4.

(iii) If n is divisible by 9, then it is divisible by 27.

OR

(i) Why does the argument "the number of factors is odd, so a factor pair repeats" prove only the converse of 'perfect squares have an odd number of factors'? **5**

(ii) How can the argument be extended to prove the proposition itself?

39. Three coins are tossed together. **5**

(i) List the sample space.

(ii) Find the probability of exactly two heads.

(iii) Find the probability of at least one head.

(iv) Find the probability that all three coins show the same face.

OR

(i) Explain subjective probability with an example. **5**

(ii) Explain how an objective estimate of the chance of rain can be made from data.

(iii) Why is rain called a random event?

40. Convert into $\frac{p}{q}$: **5**

(i) 12.6

(ii) 0.0120 (terminating)

(iii) 3.052 (terminating)

(iv) $0.\overline{23}$

(v) $2.0\overline{5}$.

OR

(i) Simplify $\left(\frac{1}{4} - \frac{5}{12}\right) \div \frac{2}{3}$. **5**

(ii) Find the sum and the product of $\frac{3}{5}$ and $\frac{7}{10}$.

END OF QUESTION PAPER **Total: 80 Marks**

MASTERMIND